5.2 Pair of Angles formed by Intersecting Lines

We are going to study related angles such as adjacent angles, linear pair of angles and vertically opposite angles.

V8U1B

5.2.1. Adjacent angles

The teacher shows a picture of sliced orange with angles marked on it.

Read the conversation between the teacher and students.

Teacher : How many angles are marked on the picture? Can you name them?
Kavin : Three angles are marked on the picture. They are $\angle AOC, \angle AOB$ and $\angle BOC$
Teacher : Which are the angles seen next to each other?
Thoorigai : Angles such as $\angle AOB$ and $\angle BOC$ are next to each other.
Teacher : How many vertices are there?
Mugil : There is only one common vertex.
Teacher : How many arms are there? Name them.
Amudhan : There are three arms. They are $\overrightarrow{OA}, \overrightarrow{OB}$ and $\overrightarrow{OC}$
Teacher : Is there any common arm for $\angle AOB$ and $\angle BOC$?
Oviya : Yes. $\overrightarrow{OB}$ is the common arm for $\angle AOB$ and $\angle BOC$.
Teacher : What can you say about the arms $\overrightarrow{OA}$ and $\overrightarrow{OC}$?
Kavin : They lie on the either side of the common arm $\overrightarrow{OB}$.
Teacher : Are the interiors of $\angle AOB$ and $\angle BOC$ overlapping?
Mugil : No. Their interiors are not overlapping.
Teacher : Hence the two angles, $\angle AOB$ and $\angle BOC$ have one common vertex (O), one common arm ($\overrightarrow{OB}$), other two arms ($\overrightarrow{OA}$ and $\overrightarrow{OC}$) lie on either side of the common arm and their interiors do not overlap.

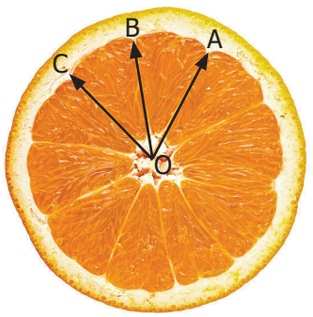


Fig.5.1

Such pair of angles $\angle AOB$ and $\angle BOC$ are called adjacent angles.

So, two angles which have a common vertex and a common arm, whose interiors do not overlap are called adjacent angles.

Now observe the Fig.5.2 in which angles are named $\angle 1$, $\angle 2$ and $\angle 3$.

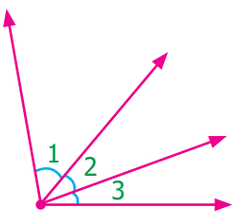


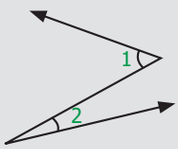
Fig.5.2

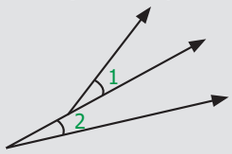
It can be observed that there are two pairs of adjacent angles such as $\angle 1$, $\angle 2$ and $\angle 2$, $\angle 3$. Then what about the pair of angles $\angle 1$ and $\angle 3$?

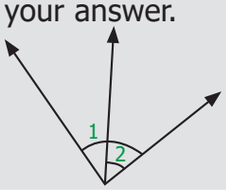
They are not adjacent because this pair of angles have a common vertex but they do not have a common arm as $\angle 2$ is in between $\angle 1$ and $\angle 3$. Also interiors of $\angle 1$ and $\angle 3$ do not overlap. Since the pair of angles does not satisfy one among the three conditions they are not adjacent.

Think

In each of the following figures, observe the pair of angles that are marked as $\angle 1$ and $\angle 2$. Do you think that they are adjacent pairs? Justify your answer.







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